

Python Web & JSON Basics

Topics:

API, REST API, JSON, Parsing, Requests

What is an API?

- ▶ API = Application Programming Interface
- ▶ Allows software systems to communicate with each other
- ▶ It is a software intermediary that allows two or more applications to talk to each other
- ▶ Web API: An API which is interface for web
- ▶ Example: A weather app using a weather API to get temperature data

API Example in Python

- ▶ `import requests`
- ▶ `response = requests.get('https://api.weatherapi.com/v1/current.json?key=API_KEY&q=London')`
- ▶ `data = response.json()`
- ▶ `print(data['current']['temp_c'])`

What is a REST API?

- ▶ REST = Representational State Transfer
- ▶ Its is an architectural guideline to develop web API.
- ▶ Uses standard HTTP methods (GET, POST, PUT, DELETE)
- ▶ Resources accessed via URLs (e.g., /books/1)
- ▶ Stateless, uses JSON or XML for data exchange

REST API Example Endpoints

- ▶ GET /books - list of books
- ▶ POST /books - add a new book
- ▶ PUT /books/1 - update book with ID 1
- ▶ DELETE /books/1 - delete book with ID 1

What is JSON?

- ▶ JSON = JavaScript Object Notation
- ▶ Lightweight, readable format for data exchange
- ▶ Used in REST APIs
- ▶ `{"name": "Asif", "age": 30, "skills": ["Python", "Django"]}`

Parsing JSON in Python

- ▶ `import json`
- ▶ `json_string = '{"name": "Asif", "age": 30}'`
- ▶ `data = json.loads(json_string)`
- ▶ `print(data['name'])`

Parsing JSON from a File

- ▶ `with open('data.json') as f:`
- ▶ `data = json.load(f)`
- ▶ `print(data['age'])`

Creating and Writing JSON

- ▶ `person = {'name': 'Asif', 'age': 30}`
- ▶ `json_string = json.dumps(person, indent=4)`
- ▶ `with open('person.json', 'w') as file:`
- ▶ `json.dump(person, file, indent=4)`

Using requests in Python

- ▶ Install with: `pip install requests`
- ▶ Use to send HTTP requests: GET, POST, PUT, DELETE

GET Request Example

- ▶ `response = requests.get('https://api.example.com/data')`
- ▶ `data = response.json()`
- ▶ `print(data)`

POST Request Example

- ▶ `payload = {'title': 'My Post', 'body': 'Test', 'userId': 1}`
- ▶ `response = requests.post(url, json=payload)`
- ▶ `print(response.json())`

PUT and DELETE Request Examples

- ▶ PUT: `requests.put(url, json=data)`
- ▶ DELETE: `requests.delete(url)`

Response Details

- ▶ `response.status_code` - HTTP status
- ▶ `response.headers` - headers
- ▶ `response.text` - raw response body